Teaching with the Population Laboratories

Routes, timings and objectives for all eleven rooms

Every room has a printable guide of its own, linked from the room's *Start here* panel. Each guide opens with learning objectives and suggested routes, then gives a **core path** sized for a BSc practical and an **extension** for MSc students. Every step asks for a prediction before the run and an explanation after it, and every guide ends with discussion questions and a page of notes for checking the reasoning. The table below is the whole set at a glance; times are minutes of active work, not counting a class discussion.

ROOM	WHAT IT IS FOR	CORE BSC	EXT. MSC
PART I — A POPULATION ON ITS OWN			
01 **The Growth Room** guides/01-growth-room.pdf	With nothing pushing back, growth is exponential; chance still drives small populations extinct.	31	+28
02 **The Crowding Room** guides/02-crowding-room.pdf	Density dependence bends exponential growth into the S-shaped logistic; K is an attractor, growth peaks at K/2, and an Allee effect adds a threshold.	32	+28
PART II — WHERE THE CEILING COMES FROM			
03 **The Resource Room** guides/03-resource-room.pdf	K emerges from a resource budget; consumers draw the resource down to R* = m/(e·a), which no amount of supply can change.	33	+30
04 **The Overshoot Room** guides/04-overshoot-room.pdf	One generation of delay turns self-regulation into overshoot: damped oscillations, cycles and deterministic chaos, depending on r and on the shape of competition.	35	+30
PART III — A SECOND SPECIES AT THE SAME TABLE			
05 **The Neighbours Room** guides/05-neighbours-room.pdf	Two competitors give four outcomes; coexistence needs each species to limit itself more than its rival, and growth rates never decide who wins.	34	+30
06 **The Coexistence Room** guides/06-coexistence-room.pdf	On one resource the lower R* wins whatever the starting numbers; with two resources species coexist when each is limited by what it eats most.	35	+30
PART IV — BEING EATEN			
07 **The Predator Room** guides/07-predator-room.pdf	Predator and prey cycle with the predator lagging; a prey ceiling damps the cycles, a saturating predator can destabilise them, and enrichment feeds the level above.	36	+28
08 **The Keystone Room** guides/08-keystone-room.pdf	A predator that presses hardest on the competitive winner lets the loser persist (keystone predation); a shared predator can exclude a species with no competition at all (apparent competition).	35	+30
PART V — MANY SPECIES AT ONCE			
09 **The Diversity Room** guides/09-diversity-room.pdf	Diversity is richness and evenness; indices weigh rare species differently, samples underestimate richness, and neutral drift drains diversity unless immigration replaces it.	36	+30
PART VI — LIFE SPREAD ACROSS SPACE			

ROOM	WHAT IT IS FOR	CORE BSC	EXT. MSC
10 **The Turnover Room** guides/10-turnover-room.pdf	Regional diversity is local diversity times turnover ($\gamma = \alpha \times \beta$); drift between isolated sites creates turnover, immigration erases it.	35	+28
11 **The Metapopulation Room** guides/11-metapopulation-room.pdf	A network of patches persists although every local population goes extinct; occupancy settles at $p^* = 1 - h - e/c$, and habitat loss lowers it one for one.	34	+28
All rooms, minutes		**376**	**+320**

HOW THE GUIDES ARE MEANT TO BE USED

Students write a prediction before they press anything, then explain whether the run matched. Several rooms are deterministic — the same settings always give the same curve — so a wrong prediction cannot be blamed on chance, which makes them especially good for this.

- **In class:** print the core-path pages; hold back the last page (*Checking your reasoning*) until the discussion, or print without it.
- **Independently:** the whole guide works as self-study, with the last page as the check.
- **For assessment:** the prediction–explanation pairs and the worksheet questions (tagged BSc or MSc) make short written tasks; the extension steps, most of which ask for a calculation before the run, make good MSc problem-sheet questions.

A BSC COURSE IN SIX PRACTICALS OF ABOUT TWO HOURS

- **Practical 1 — a population on its own:** Growth (01) and Crowding (02).
- **Practical 2 — where the ceiling comes from:** Resource (03) and Overshoot (04).
- **Practical 3 — competition:** Neighbours (05) and Coexistence (06).
- **Practical 4 — predation:** Predator (07) and Keystone (08).
- **Practical 5 — describing communities:** Diversity (09), with a short field sample if the timetable allows.
- **Practical 6 — landscapes:** Turnover (10) and Metapopulation (11).

AN MSC MODULE

- Set each room's core path as preparation before class, with the *Checking your reasoning* page.
- Spend contact time on the extension steps: most ask for an equilibrium or threshold to be calculated before the run and checked against it.
- The MSc questions of Coexistence, Keystone and Metapopulation make a good written assignment on what regulates populations and communities.

SINGLE SESSIONS

- **What stops a population growing? (90 minutes):** Growth (01), Crowding (02) and Resource (03) steps 1–2.
- **Cycles two ways (one hour):** Overshoot (04) steps 1–4 and Predator (07) steps 1–3, then question 3 of either.
- **Coexistence (two hours):** Neighbours (05), Coexistence (06) and Keystone (08) core paths.
- **Conservation in landscapes (90 minutes):** Turnover (10) step 4 and Metapopulation (11) core path.